4.1.3 User Cases

UC_01: User logs in

UC Name	User logs in
Dependency	None
Actors	Admin, Agent, Property Inspector, Property Owner, Buyer/Tenant
Preconditions	User has a registered and active account in the system.
Description of Main Sequence	1. User navigates to the TrustEstate login page. 2. User enters their email and password. 3. User clicks the Login button. 4. System validates the credentials. 5. System identifies the user role. 6. System redirects the user to their role-specific dashboard.
Description of Alternative Sequence	1. Invalid credentials: System displays an error message and prompts the user to re-enter credentials. 2. Forgotten password: User clicks the forgot password link and requests a password reset. 3. Account suspended/deactivated: System notifies the user that their account is inactive and to contact the Admin.
Non-functional requirements	1. Login process should complete within 3 seconds. 2. Passwords must be stored in encrypted format. 3. System must log all failed login attempts. 4. System must lock or flag an account after a defined number of consecutive failed login attempts. 5. Session tokens must expire after a defined period of inactivity.
Postconditions	1. User is authenticated and has access to their role-specific dashboard. 2. System logs the successful login.

UC_02: Review pending verifications

UC Name	Review pending verifications
Dependency	US_01 (User logs in)
Actors	Admin
Preconditions	1. Admin is authenticated. 2. At least one Agent or Property Inspector account is awaiting verification.
Description of Main Sequence	1. Admin navigates to the Pending Verifications section of the dashboard. 2. System displays a list of all pending Agent and Property Inspector accounts. 3. Admin selects a pending account and reviews the submitted professional details and agency affiliation. 4. If approved: System activates the account, assigns a Verified badge, and notifies the user. 5. If rejected: System keeps the account inactive and notifies the user with a documented reason. 6. System logs the decision within the 72-hour review window.
Description of Alternative Sequence	No pending accounts: System displays a message indicating no accounts are awaiting verification.
Non-functional requirements	1. The pending verifications list must load within 3 seconds. 2. The system must ensure that no verification decision can be made without a documented reason in the case of rejection. 3. All verification decisions must be timestamped and logged for audit purposes. 4. The 72-hour review window must be enforced and tracked by the system. 5. Notifications to the user upon approval or rejection must be delivered within 1 minute of the Admin's decision.
Postconditions	1. The account is either activated with a Verified badge or rejected with a documented reason. 2. The user is notified of the outcome.

UC_03: Monitor and manage listings

UC Name	Monitor and manage listings
Dependency	US_01 (User logs in)
Actors	Admin
Preconditions	1. Admin is authenticated. 2. At least one active listing exists on the platform.
Description of Main Sequence	1. Admin navigates to the Listings Management section of the dashboard. 2. System displays all active, flagged, and suspended listings. 3. Admin reviews a listing for policy compliance. 4. Admin selects an action: flag, suspend, or remove the listing. 5. Admin enters a documented reason for the action. 6. System applies the action and notifies the assigned Agent and Property Owner with the documented reason.
Description of Alternative Sequence	1. No listings found: System displays a message indicating no listings are currently on the platform. 2. Admin cancels action: No changes are applied and the listing remains unchanged.
Non-functional requirements	1. The listings management section must load and display all listings within 3 seconds. 2. Any action applied to a listing must be reflected in the system immediately. 3. Notifications to the assigned Agent and Property Owner must be delivered within 1 minute of the Admin's action. 4. All Admin actions on listings must be logged with a timestamp and documented reason for audit purposes. 5. The system must support filtering and sorting of listings by status, date, and assigned Agent.
Postconditions	1. The selected action is applied to the listing. 2. The assigned Agent and Property Owner are notified with the documented reason.

UC_04: Monitor inspection reports

UC Name	Monitor inspection reports
Dependency	US_01 (User logs in)
Actors	Admin
Preconditions	1. Admin is authenticated. 2. At least one inspection report has been submitted on the platform.
Description of Main Sequence	1. Admin navigates to the Inspection Reports section of the dashboard. 2. System displays all inspection reports submitted on the platform. 3. Admin selects a report and reviews it for quality and compliance with platform standards. 4. If the report appears fraudulent or non-compliant: Admin flags or removes it. 5. System records the action and notifies the relevant Agent and Property Inspector.
Description of Alternative Sequence	1. No reports found: System displays a message indicating no inspection reports have been submitted. 2. Admin cancels action: No changes are applied and the report remains unchanged.
Non-functional requirements	1. The inspection reports section must load within 3 seconds. 2. Any flagging or removal action applied to a report must be reflected in the system immediately. 3. All Admin actions on inspection reports must be logged with a timestamp and documented reason for audit purposes. 4. Notifications to the relevant Agent and Property Inspector must be delivered within 1 minute of the Admin's action. 5. The system must ensure that flagged or removed reports are retained in a system log and not permanently deleted without a record.
Postconditions	1. The selected action is recorded and applied to the report. 2. Relevant parties are notified where applicable.

UC_05: Manage user accounts

UC Name	Manage user accounts
Dependency	US_01 (User logs in)

Actors	Admin
Preconditions	1. Admin is authenticated. 2. At least one registered user account exists in the system.
Description of Main Sequence	1. Admin navigates to the User Management section of the dashboard. 2. System displays all registered users, filterable by role and account status. 3. Admin selects a user account to review. 4. Admin chooses to deactivate, suspend, or reactivate the account. 5. Admin enters a documented reason for the action. 6. System applies the status change and notifies the affected user with the documented reason.
Description of Alternative Sequence	1. No users found for selected filter: System displays a message indicating no matching accounts. 2. Admin cancels action: No changes are applied and the account status remains unchanged.
Non-functional requirements	1. The user management section must load within 3 seconds. 2. Any account status change must be applied and reflected in the system immediately. 3. Notifications to the affected user must be delivered within 1 minute of the Admin's action. 4. All account status changes must be logged with a timestamp and documented reason for audit purposes. 5. A deactivated or suspended user must be immediately prevented from logging in upon the status change taking effect.
Postconditions	1. The user account status is updated. 2. The affected user is notified with the documented reason.

UC_06: Manage disputes

UC Name	Manage disputes
Dependency	US_01 (User logs in)
Actors	Admin
Preconditions	1. Admin is authenticated. 2. At least one open dispute has been submitted by a platform user.
Description of Main Sequence	1. Admin navigates to the Disputes section of the dashboard. 2. System displays all open disputes submitted by platform users. 3. Admin selects a dispute and reviews the listing details, offer history, inspection reports, and communications between parties. 4. Admin communicates with both parties through the platform messaging system. 5. If the dispute is resolvable: Admin marks it as Resolved with a documented outcome and notifies all involved parties. 6. If one party is unresponsive: Admin temporarily suspends the transaction and notifies all involved parties.
Description of Alternative Sequence	1. No open disputes: System displays a message indicating no disputes are currently open.

Non-functional requirements	1. The disputes section must load and display all open disputes within 3 seconds. 2. All dispute-related communications through the platform must be logged and retained for audit purposes. 3. Any transaction suspension applied by the Admin must take effect immediately and notify all involved parties within 1 minute. 4. Dispute resolution outcomes must be documented and timestamped in the system. 5. The system must ensure that all parties involved in a dispute have equal and simultaneous access to any communications directed at them.
Postcondition	1. The dispute is either marked as Resolved with a documented outcome, or escalated with a temporary transaction suspension. 2. All involved parties are notified.

UC_07: View analytics dashboard

UC Name	View analytics dashboard
Dependency	US_01 (User logs in)
Actors	Admin
Preconditions	Admin is authenticated.
Description of Main Sequence	1. Admin navigates to the Analytics Dashboard. 2. System displays real-time data including active listings, flagged listings, pending verifications, ongoing transactions, registered users per role, and open disputes. 3. Admin reviews the platform-wide metrics and activity. 4. Admin may filter data by category for a more focused view.
Description of Alternative Sequence	1. Insufficient data: System displays a message indicating no data is available for the selected category or filter.
Non-functional requirements	1. The analytics dashboard must load and display real-time data within 3 seconds. 2. Data displayed on the dashboard must reflect the current state of the platform with no more than a 1-minute delay. 3. The dashboard must remain accessible to the Admin at all times without service interruption. 4. The system must support filtering of analytics data by category without requiring a full page reload.
Postconditions	1. Admin has reviewed the current platform-wide analytics. 2. Data displayed remains unchanged by this action.

UC_08: Property Inspector registers

UC Name	Property inspector registers
Dependency	None
Actors	Property Inspector
Preconditions	1. The user does not have to have an account on the platform. 2. The registration page is accessible and operational.
Description of Main Sequence	1. User navigates to the TrustEstate public registration page. 2. User selects Property Inspector as their professional type. 3. User fills in the required personal and professional details and submits the form. 4. System creates the account in Pending Verification status. 5. System notifies the Admin of the new pending account. 6. User receives a confirmation that their account is awaiting Admin approval before activation.
Description of Alternative Sequence	1. Email already registered: System displays an error and prompts the user to use a different email or log in. 2. Invalid or incomplete form data: System highlights the problematic fields and prompts the user to correct them before resubmitting.
Non-functional requirements	1. The registration form must load within 2 seconds. 2. Form validation must execute in real time, highlighting errors before submission. 3. The account creation process must complete within 3 seconds of a valid form submission. 4. The Admin notification of the new pending account must be delivered within 1 minute of successful registration. 5. All submitted registration data must be stored securely and encrypted where applicable.
Postconditions	1. A new Property Inspector account is created in Pending Verification status. 2. The Admin is notified to review the account within 72 hours.

UC_09: Property Inspector receives Verified badge

UC Name	Property Inspector receives Verified badge
Dependency	US_08 (Property Inspector registers) US_02 (Review pending verifications)
Actors	Property Inspector
Preconditions	1. The Property Inspector's account has been approved by the Admin.
Description of Main Sequence	1. System notifies the Property Inspector that their account has been approved by the Admin. 2. Property Inspector logs in following UC_01. 3. Inspector navigates to their profile page. 4. System displays the Verified badge on the inspector's profile. 5. Inspector is now eligible to be assigned to inspections by Agents.
Description of Alternative Sequence	1. Account rejected: Inspector receives a notification with the documented reason for rejection and their account remains inactive.
Non-functional requirements	1. The approval notification must be delivered to the Property Inspector within 1 minute of the Admin's decision. 2. The Verified badge must appear on the inspector's profile immediately upon account activation. 3. The system must ensure that only Admin-approved accounts receive the Verified badge - no manual override by the inspector is possible.
Postconditions	1. The Property Inspector's account is fully active and displays a Verified badge. The inspector is eligible for assignment by Agents.

UC_10: View assigned inspections

UC Name	View assigned inspections
Dependency	US_01 (User logs in)
Actors	Property Inspector
Preconditions	1. Property Inspector is authenticated and their account is active and verified. 2. At least one inspection has been assigned to them.
Description of Main Sequence	1. Inspector navigates to the Assigned Inspections section of the dashboard. 2. System displays all inspections currently assigned to the inspector. 3. Each entry shows the property details, scheduled inspection date, and the assigned Agent's contact information. 4. Inspector selects an inspection entry to view its full details.
Description of Alternative Sequence	1. No assigned inspections: System displays a message indicating no inspections are currently assigned.
Non-functional requirements	1. The assigned inspections list must load within 3 seconds. 2. All inspection entries must display up-to-date information reflecting the most recent status and assignment details. 3. The system must ensure that the inspector can only view inspections explicitly assigned to them and no others.
Postconditions	1. Inspector has successfully viewed their assigned inspections and can proceed with the relevant inspection workflow.

UC_11: Update inspection status

UC Name	Update inspection status
Dependency	US_01 (User logs in) US_10 (View assigned inspections)
Actors	Property Inspector
Preconditions	1. Property Inspector is authenticated. An inspection is assigned to them with a status of Scheduled or In Progress. 2. The update is performed on the day of the inspection.
Description of Main Sequence	1. Inspector navigates to the relevant assigned inspection on the day of the inspection. 2. Inspector updates the status from Scheduled to In Progress. 3. System records the status change and notifies the assigned Agent. 4. After completing the physical inspection, inspector updates the status from In Progress to Completed. 5. System records the completion and prompts the inspector to submit the inspection report.
Description of Alternative Sequence	1. Inspector attempts to update status before the scheduled date: System prevents the update and displays an informational message.

Non-functional requirements	1. Status updates must be applied and reflected in the system immediately upon submission. 2. Notifications to the assigned Agent upon a status change must be delivered within 1 minute. 3. The system must enforce that the status can only be updated in the correct sequence: Scheduled → In Progress → Completed. Skipping steps must not be permitted. 4. The system must prevent status updates to In Progress before the scheduled inspection date.
Postconditions	1. The inspection status is updated to Completed. The system prompts the inspector to proceed with submitting the inspection report.

US_12: Submit inspection report

UC Name	Submit inspection report
Dependency	US_01 (User logs in) US_11 (Update inspection status)
Actors	Property Inspector
Preconditions	1. Property Inspector is authenticated. 2. The inspection status has been updated to Completed. 3. The report has not yet been submitted.
Description of Main Sequence	1. Inspector navigates to the completed inspection and opens the report submission form. 2. Inspector fills in findings for each of the four fixed categories: Structural Integrity, Plumbing, Electrical, and Safety. 3. For each category, inspector enters written findings, a pass/fail rating, a severity classification (Minor, Moderate, or Critical), and uploads supporting photos. 4. Inspector reviews all entered data for accuracy. 5. Inspector submits the report. 6. System saves and locks the report — it cannot be modified or resubmitted once formally submitted.
Description of Alternative Sequence	1. Incomplete report: System highlights missing required fields and prevents submission until all fields are completed. 2. Upload failure: System notifies the inspector of the photo upload failure and prompts a retry.
Non-functional requirements	1. The report submission form must load within 2 seconds. 2. Photo uploads per category must support common image formats and must not exceed a defined file size limit per upload. 3. The report submission process must complete within 5 seconds of a valid form submission. 4. Once submitted, the report must be immediately and permanently locked - no edit or resubmission must be possible under any circumstance. 5. All submitted report data must be stored securely and backed up.
Postconditions	1. The inspection report is formally submitted and permanently locked. 2. The system prompts the inspector to assign the final verdict.

UC_13: Assign final verdict

UC Name	Assign final verdict
Dependency	US_01 (User logs in) US_12 (Submit inspection report)
Actors	Property Inspector
Preconditions	1. Property Inspector is authenticated. 2. The inspection report has been submitted. 3. A final verdict has not yet been assigned.
Description of Main Sequence	1. Inspector navigates to the submitted inspection report. 2. Inspector selects the final verdict: Passed, Passed with Conditions, or Failed. 3. Inspector confirms the verdict submission. 4. System automatically sends notifications to the Buyer/Tenant, Agent, and Property Owner with the verdict outcome. 5. System makes the full inspection report accessible to the Buyer/Tenant and Property Owner. 6. Report is permanently locked — no modifications or resubmissions are permitted.
Description of Alternative Sequence	1. Inspector attempts to modify a submitted verdict: System prevents the action and displays a message indicating the report is locked.
Non-functional requirements	1. The verdict submission must be processed and recorded by the system within 3 seconds. 2. Notifications to the Buyer/Tenant, Agent, and Property Owner must be delivered within 1 minute of verdict submission. 3. The full inspection report must become accessible to authorized parties immediately upon verdict submission. 4. The system must ensure that the verdict and report are permanently locked upon submission — no modifications are permitted under any circumstance.
Postconditions	1. The final verdict is recorded and the report is permanently locked. 2. Notifications are sent to the Buyer/Tenant, Agent, and Property Owner. 3. The full report becomes accessible to authorized parties.

UC_14: View inspection history

UC Name	View inspection history
Dependency	US_01 (User logs in)
Actors	Property Inspector
Preconditions	1. Property Inspector is authenticated. 2. At least one inspection has been previously completed and submitted by the inspector.
Description of Main Sequence	1. Inspector navigates to the Inspection History section of the dashboard. 2. System displays all inspections previously completed by the inspector. 3. Inspector selects an entry to view the full submitted report, including findings, severity classifications, photos, and the final verdict. 4. Inspector may use the history as a reference but cannot modify any submitted report.
Description of Alternative Sequence	1. No completed inspections: System displays a message indicating no inspection history is available.
Non-functional requirements	1. The inspection history list must load within 3 seconds. 2. The system must ensure that the inspector can only access their own inspection history and not that of other inspectors. 3. All historical report data must be stored securely and remain intact and unmodifiable.
Postconditions	1. Inspector has successfully reviewed their inspection history. 2. No changes are made to any submitted report.

UC_15: Agent registers

UC Name	Agent registers
Dependency	None
Actors	Agent
Preconditions	1. The user does not already have an account on the platform. 2. The registration page is accessible and operational.

Description of Main Sequence	1. User navigates to the TrustEstate public registration page. 2. User selects Agent as their professional type. 3. User fills in personal details and provides agency affiliation, selecting either Independent or entering their Agency Name. 4. System creates the account in Pending Verification status. 5. System notifies the Admin of the new pending Agent account. 6. User receives a confirmation that their account is awaiting Admin approval before activation.
Description of Alternative Sequence	1. Email already registered: System displays an error and prompts the user to use a different email or log in. 2. Invalid or incomplete form data: System highlights the problematic fields and prompts the user to correct them before resubmitting.
Non-functional requirements	1. The registration form must load within 2 seconds. 2. Form validation must execute in real time, highlighting errors before submission. 3. The account creation process must complete within 3 seconds of a valid form submission. 4. The Admin notification of the new pending Agent account must be delivered within 1 minute of successful registration. 5. All submitted registration data must be stored securely and encrypted where applicable.
Postconditions	1. A new Agent account is created in Pending Verification status. 2. The Admin is notified to review the account within 72 hours.

UC_16: Agent receives Verified badge

UC Name	Agent receives Verified badge
Dependency	US_15 (Agent registers) US_02 (Review pending verifications)
Actors	Agent
Preconditions	1. The Agent's account has been approved by the Admin.
Description of Main Sequence	1. System notifies the Agent that their account has been approved by the Admin. 2. Agent logs in following UC_01. 3. Agent navigates to their profile page. 4. System displays the Verified badge on the Agent's profile. 5. Agent gains full access to the agent dashboard and is now visible to Property Owners when selecting an Agent for a listing assignment.
Description of Alternative Sequence	1. Account rejected: Agent receives a notification with the documented reason for rejection and their account remains inactive.
Non-functional requirements	1. The approval notification must be delivered to the Agent within 1 minute of the Admin's decision. 2. The Verified badge must appear on the Agent's profile immediately upon account activation. 3. The Agent must become visible to Property Owners in the Agent selection list immediately upon account activation.

	4. The system must ensure that only Admin-approved Agent accounts receive the Verified badge and appear in the selection list.
Postconditions	1. The Agent's account is fully active and displays a Verified badge. 2. The Agent is visible to Property Owners and eligible to receive listing assignments.

UC_17: Accept or decline listing assignment

UC Name	Accept or decline listing assignment
Dependency	US_01 (User logs in) US_24 (Create and submit listing)
Actors	Agent
Preconditions	1. Agent is authenticated and their account is active and verified. 2. A Property Owner has selected this Agent for a listing assignment.
Description of Main Sequence	1. System notifies the Agent of a new listing assignment request from a Property Owner. 2. Agent navigates to the notification and reviews the listing details. 3. If the Agent accepts: System assigns the Agent to the listing and updates its status to Pending Agent Review. 4. If the Agent declines: System notifies the Property Owner to select a different Agent.
Description of Alternative Sequence	1. Agent does not respond: The assignment remains pending until the Agent takes action or the Property Owner selects a different Agent.
Non-functional requirements	1. The assignment notification must be delivered to the Agent within 1 minute of the Property Owner's submission. 2. The Agent's acceptance or decline must be processed and reflected in the listing status immediately. 3. The Property Owner must be notified of the Agent's decision within 1 minute of the action being taken.
Postconditions	1. If accepted: The Agent is assigned to the listing and its status is updated to Pending Agent Review. 2. If declined: The Property Owner is notified to select a different Agent.

UC_18: Review and approve listings

UC Name	Review and approve listings
Dependency	US_01 (User logs in) US_17 (Accept or decline listing assignment)
Actors	Agent
Preconditions	1. Agent is authenticated. 2. At least one listing assigned to the Agent has a status of Pending Agent Review.
Description of Main Sequence	1. Agent navigates to assigned listings and selects one with Pending Agent Review status. 2. Agent reviews the full listing details: title, description, location, price, listing type, and photos. 3. If corrections are needed: Agent requests corrections from the Property Owner with a documented note; listing status changes to Corrections Requested. 4. Property Owner edits and resubmits the listing; Agent reviews it again from step 2. 5. If the listing meets all quality standards: Agent approves it; system sets the status to Active and makes it publicly visible. 6. System notifies the Property Owner that their listing is now live.
Description of Alternative Sequence	1. No listings pending review: System displays a message indicating no listings are awaiting review. 2. Agent cancels review: No changes are applied and the listing status remains unchanged.
Non-functional requirements	1. The listing details page must load within 3 seconds. 2. Any status change applied by the Agent (approval or corrections requested) must be reflected in the system immediately. 3. Notifications to the Property Owner upon approval or correction request must be delivered within 1 minute. 4. Upon approval, the listing must become publicly visible to all Buyer/Tenant users immediately. 5. All Agent actions on listings must be logged with a timestamp for audit purposes.
Postconditions	1. If approved: Listing status is set to Active and it is publicly visible to all Buyer/Tenant users. Property Owner is notified. 2. If corrections requested: Listing status is set to Corrections Requested and Property Owner is notified with a documented reason.

UC_19: Manage offer lifecycle

UC Name	Manage offer lifecycle
Dependency	US_01 (User logs in) US_35 (Submit offer)
Actors	Agent

Preconditions	1. Agent is authenticated. 2. At least one offer has been submitted on a listing assigned to the Agent.
Description of Main Sequence	1. Agent receives a notification when a Buyer/Tenant submits an offer on an assigned listing. 2. Agent navigates to the offer and reviews the proposed price and optional message. 3. Agent selects an action on behalf of the Property Owner: Accept, Decline, or Counter. 4. If accepted: System updates the offer status to Accepted and notifies the Buyer/Tenant; listing status changes to Under Offer. 5. If declined: System updates the offer status to Declined and notifies the Buyer/Tenant. 6. If countered: Agent submits a revised price and sets a response deadline; system notifies the Buyer/Tenant. 7. Negotiation continues for a maximum of 3 rounds; if unresolved, the offer expires and the listing returns to Active status.
Description of Alternative Sequence	1. Negotiation reaches 3 rounds without resolution: System automatically expires the offer and returns the listing to Active status, notifying the Buyer/Tenant.
Non-functional requirements	1. Offer response notifications to the Buyer/Tenant must be delivered within 1 minute of the Agent's action. 2. All offer actions must be processed and reflected in the system immediately. 3. The system must enforce the maximum of 3 negotiation rounds automatically, expiring the offer without requiring manual Agent intervention. 4. The response deadline set by the Agent for a counter-offer must be enforced by the system, automatically expiring the offer if the deadline passes without a response. 5. All offer lifecycle events must be logged with timestamps for audit and dispute resolution purposes.
Postconditions	1. If accepted: Offer status is Accepted and listing status is Under Offer. 2. If declined: Offer status is Declined and the listing remains Active. 3. If countered: Buyer/Tenant is notified and the negotiation round count increments. 4. If expired: Offer is closed and listing returns to Active status.

UC_20 Assign Property Inspector

UC Name	Assign Property Inspector
Dependency	US_01 (User logs in) US_19 (Manage offer lifecycle)
Actors	Agent
Preconditions	1. Agent is authenticated. 2. An offer on one of the Agent's assigned listings has been accepted. 3. At least one verified Property Inspector is available on the platform.
Description of Main Sequence	1. Agent navigates to the listing's transaction management panel after an offer is accepted. 2. System displays a list of all available verified Property Inspectors. 3. Agent selects an Inspector and schedules the inspection date. 4. System creates an inspection record and notifies the Inspector with the property details and scheduled date. 5. Buyer/Tenant and Property Owner receive a notification that an inspection has been scheduled. 6. If the assigned Inspector becomes unavailable before the inspection: Agent selects a replacement Inspector. 7. System updates the assignment and notifies all relevant parties.
Description of Alternative Sequence	1. No verified Inspectors available: System displays a message indicating no verified Inspectors are currently available. 2. Agent must wait until an Inspector becomes available.
Non-functional requirements	1. The list of available verified Property Inspectors must load within 3 seconds. 2. The inspection assignment must be created and reflected in the system immediately upon the Agent's confirmation. 3. Notifications to the assigned Property Inspector, Buyer/Tenant, and Property Owner must be delivered within 1 minute of the assignment. 4. The system must ensure that only verified Property Inspectors appear in the selection list - unverified inspectors must not be assignable.
Postconditions	1. An inspection record is created and assigned to the selected verified Property Inspector. 2. All relevant parties are notified of the scheduled inspection.

UC_21: Close transaction

UC Name	Close transaction
Dependency	US_1 (User logs in) US_13 (Assign final verdict)
Actors	Agent
Preconditions	1. Agent is authenticated. 2. The offer is accepted, the inspection is completed and the report has been submitted, and there are no pending disputes on the listing.
Description of Main Sequence	1. Agent verifies that all conditions are met: offer accepted, inspection completed, no pending disputes. 2. Agent navigates to the listing's transaction panel and selects Close Transaction. 3. Agent confirms the closure. 4. System marks the transaction as Closed, archives the listing, and notifies the Buyer/Tenant, Property Owner, and Property Inspector.
Description of Alternative Sequence	1. Conditions not met: System prevents the Agent from closing the transaction and displays a message indicating which conditions remain outstanding.
Non-functional requirements	1. The transaction closure must be processed and the listing archived immediately upon the Agent's confirmation. 2. Notifications to all relevant parties must be delivered within 1 minute of the closure. 3. The system must enforce that a transaction cannot be closed unless all required conditions are met - accepted offer, completed inspection, and no pending disputes. 4. All transaction closure events must be logged with a timestamp for audit purposes.
Postconditions	1. The transaction is marked as Closed. 2. The listing is archived and no longer publicly visible. 3. All relevant parties are notified of the closure.

UC_22: Message Buyer/Tenant

UC Name	Message Buyer/Tenant
Dependency	US_01 (User logs in) US_19 (Manage offer lifecycle)
Actors	Agent
Preconditions	1. Agent is authenticated. 2. At least one Buyer/Tenant has submitted an offer on a listing assigned to the Agent.

Description of Main Sequence	1. Agent navigates to the messaging section of the platform. 2. Agent selects the Buyer/Tenant who has submitted an offer on one of their assigned listings. 3. Agent composes and sends a message through the platform's internal messaging system. 4. System delivers the message and notifies the Buyer/Tenant. 5. Buyer/Tenant may reply through the same messaging thread.
Description of Alternative Sequence	1. No eligible Buyer/Tenant to message: System displays a message indicating no offers have been submitted on the Agent's listings.
Non-functional requirements	1. Messages must be delivered to the recipient within 1 minute of being sent. 2. The messaging interface must load within 2 seconds. 3. The system must ensure that Agents can only initiate or participate in messaging threads with Buyer/Tenants who have submitted offers on their assigned listings. 4. All messages must be stored securely and retained for audit and dispute resolution purposes.
Postconditions	1. The message is delivered to the Buyer/Tenant through the platform's internal messaging system. 2. The Buyer/Tenant is notified.

UC_23: Property Owner registers

UC Name	Property Owner registers
Dependency	None
Actors	Property Owner
Preconditions	1. The user does not already have an account on the platform. 2. The registration page is accessible and operational.
Description of Main Sequence	1. User navigates to the TrustEstate public registration page. 2. User selects Property Owner as their professional type. 3. User fills in the required personal details and submits the registration form. 4. System creates the account and activates it immediately with no Admin approval required. 5. User receives a confirmation and is redirected to the Property Owner dashboard.
Description of Alternative Sequence	1. Email already registered: System displays an error and prompts the user to use a different email or log in. 2. Invalid or incomplete form data: System highlights the problematic fields and prompts the user to correct them before resubmitting.
Non-functional requirements	1. The registration form must load within 2 seconds. 2. Form validation must execute in real time, highlighting errors before submission. 3. The account creation and activation process must complete within 3 seconds of a valid form submission. 4. All submitted registration data must be stored securely and encrypted where applicable.

Postconditions	1. A new Property Owner account is created and immediately active. 2. The user is redirected to the Property Owner dashboard.

UC_24: Create and submit listing

UC Name	Create and submit listing
Dependency	US_01 (User logs in)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. At least one verified Agent is available on the platform for selection.
Description of Main Sequence	1. Owner navigates to the Create Listing section of the dashboard. 2. Owner fills in the listing details: title, description, location, asking price, listing type (sale/rent), and uploads photos. 3. Owner selects an available verified Agent from the list to manage the listing. 4. Owner reviews the listing and submits it for Agent review. 5. System creates the listing in Pending Agent Review status and notifies the selected Agent of the assignment request.
Description of Alternative Sequence	1. No verified Agents available: System displays a message indicating no verified Agents are currently available for selection. 2. Incomplete listing details: System highlights missing required fields and prevents submission until all fields are completed.
Non-functional requirements	1. The listing creation form must load within 2 seconds. 2. Photo uploads must support common image formats and must not exceed a defined file size limit per upload. 3. The listing submission process must complete within 5 seconds of a valid form submission. 4. The selected Agent must be notified of the assignment request within 1 minute of the listing submission. 5. The system must ensure that only verified Agents appear in the Agent selection list.
Postconditions	1. The listing is created in Pending Agent Review status. 2. The selected Agent is notified of the assignment request.

UC_25: Edit listing

UC Name	Edit listing
Dependency	US_01 (User logs in) US_24 (Create and submit listing)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. The listing has a status of Pending Agent Review or Corrections Requested.

Description of Main Sequence	1. Owner navigates to the listings dashboard and selects a listing in Pending Agent Review or Corrections Requested status. 2. Owner edits the required listing details: title, description, location, price, listing type, or photos. 3. Owner reviews the changes and resubmits the listing to the Agent. 4. System updates the listing status to Pending Agent Review and notifies the Agent. 5. If the listing is Active: Owner cannot edit it directly and must contact the assigned Agent through the platform messaging system.
Description of Alternative Sequence	1. Owner attempts to edit an Active listing: System prevents the edit and displays a message instructing the owner to contact the assigned Agent.
Non-functional requirements	1. The listing edit form must load within 2 seconds. 2. Any updates to the listing must be saved and reflected in the system immediately upon resubmission. 3. The assigned Agent must be notified of the resubmission within 1 minute. 4. The system must enforce that Active listings cannot be edited directly by the Property Owner — the restriction must be applied automatically.
Postconditions	1. The listing details are updated and the listing status is set to Pending Agent Review. The assigned Agent is notified to re-review the listing.

UC_26: Remove listing

UC Name	Remove listing
Dependency	US_01 (User logs in) US_24 (Create and submit listing)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. The listing exists and has no active transaction in progress.
Description of Main Sequence	1. Owner navigates to the listings dashboard and selects the listing to remove. 2. System checks whether the listing has an active transaction in progress. 3. If no active transaction: System prompts the owner to confirm the removal. 4. Owner confirms; system removes the listing from the platform.

Description of Alternative Sequence	1. Active transaction in progress: System prevents removal and notifies the owner that the listing cannot be removed while a transaction is active. 2. Owner cancels removal: No changes are applied and the listing remains unchanged.
Non-functional requirements	1. The system must check for an active transaction in real time before allowing the removal action to proceed. 2. The removal must be processed and the listing taken down immediately upon the owner's confirmation. 3. The system must enforce that listings with active transactions cannot be removed - this restriction must be applied automatically without requiring Admin intervention.
Postconditions	1. If confirmed: The listing is permanently removed from the platform. If an active transaction exists: The listing remains unchanged and the owner is informed.

UC_27: View offers on listing

UC Name	View offers on listing
Dependency	US_01 (User logs in) US_24 (Create and submit listing)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. At least one offer has been submitted on one of their listings.
Description of Main Sequence	1. Owner navigates to the listing details page and opens the Offers section. 2. System displays all submitted offers with proposed prices, optional messages, and current statuses. 3. System notifies the owner whenever a new offer is submitted or an offer status changes. 4. Owner may view the full offer history and negotiation rounds for each offer. 5. Owner cannot accept, decline, or counter offers - these actions are handled exclusively by the assigned Agent.
Description of Alternative Sequence	1. No offers submitted: System displays a message indicating no offers have been submitted on the selected listing.
Non-functional requirements	1. The offers section must load within 3 seconds. 2. Offer data displayed must reflect the most current status in real time. 3. The system must ensure that the Property Owner can only view offers on their own listings. 4. Notifications to the Property Owner for new offers or offer status changes must be delivered within 1 minute of the triggering event.
Postconditions	1. Owner has reviewed all offers and their statuses on the selected listing. No changes to offers or listing status are made by this action.

UC_28: Monitor listing statuses

UC Name	Monitor listing statuses
Dependency	US_01 (User logs in) US_24 (Create and submit listing)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. At least one listing exists under their account.
Description of Main Sequence	1. System displays all of the owner's listings with their current statuses on the dashboard upon login. 2. Owner selects a listing to view a detailed status breakdown and transaction history. 3. System sends automatic notifications to the owner at every relevant stage of the transaction.
Description of Alternative Sequence	1. No listings found: System displays a message indicating the owner has no listings yet and prompts them to create one.
Non-functional requirements	1. The listings dashboard must load within 3 seconds upon login. 2. Listing statuses displayed must reflect the most current state of each listing in real time. 3. Automatic notifications to the Property Owner at each transaction stage must be delivered within 1 minute of the triggering event. 4. The system must ensure that the Property Owner can only view their own listings on the dashboard.
Postconditions	1. Owner has reviewed the current status of their listings. No changes to any listing are made by this action.

UC_29: View Inspection report (Property Owner)

UC Name	View Inspection report
Dependency	US_01 (User logs in) US_13 (Assign final verdict)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. An inspection report has been submitted for one of their properties.
Description of Main Sequence	1. Owner receives a notification that an inspection report has been submitted for one of their properties. 2. Owner logs in following UC_01 and navigates to the listing's inspection section. 3. System displays the full inspection report including category findings (Structural Integrity, Plumbing, Electrical, Safety), pass/fail ratings, severity classifications, supporting photos, and the final verdict. 4. Owner reviews the report. 5. Owner may contact the assigned Agent to discuss the outcome through the platform messaging system. 6. Owner cannot modify the inspection report.

Description of Alternative Sequence	1. No inspection report available: System displays a message indicating no report has been submitted for the selected property yet.
Non-functional requirements	1. The inspection report must load and display in full within 3 seconds of navigation. 2. The notification to the Property Owner upon report submission must be delivered within 1 minute of the Inspector's submission. 3. The system must ensure that the Property Owner can only access inspection reports for their own properties. 4. The report must be displayed in a clear and structured format organized by the four inspection categories.
Postconditions	1. Owner has reviewed the full inspection report. 2. No modifications to the report are permitted.

UC_30: Communicate with agent

UC Name	Communicate with agent
Dependency	US_01 (User logs in) US_24 (Create and submit listing)
Actors	Property Owner
Preconditions	1. Property Owner is authenticated. 2. A listing with an assigned Agent exists under their account.
Description of Main Sequence	1. Owner navigates to the messaging section of the platform. 2. Owner selects the Agent assigned to the relevant listing. 3. Owner composes and sends a message regarding listing or transaction matters. 4. System delivers the message and notifies the Agent. 5. Agent may reply through the same messaging thread.
Description of Alternative Sequence	1. No assigned Agent available to message: System displays a message indicating no Agent is currently assigned to any of their listings.
Non-functional requirements	1. Messages must be delivered to the Agent within 1 minute of being sent. 2. The messaging interface must load within 2 seconds. 3. The system must ensure that the Property Owner can only initiate messaging with the Agent assigned to their listing. 4. All messages must be stored securely and retained for audit and dispute resolution purposes.
Postconditions	1. The message is delivered to the assigned Agent through the platform's internal messaging system. The Agent is notified.

UC_31: Buyer/Tenant registers

UC Name	Buyer/Tenant registers
Dependency	None
Actors	Buyer/Tenant
Preconditions	1. The user does not already have an account on the platform. 2. The registration page is accessible and operational.
Description of Main Sequence	1. User navigates to the TrustEstate public registration page. 2. User fills in the required personal details with no professional type selection required. 3. User submits the registration form. 4. System creates the account and activates it immediately with no Admin approval required. 5. User receives a confirmation and is redirected to the Buyer/Tenant dashboard.
Description of Alternative Sequence	1. Email already registered: System displays an error and prompts the user to use a different email or log in. 2. Invalid or incomplete form data: System highlights the problematic fields and prompts the user to correct them before resubmitting.
Non-functional requirements	1. The registration form must load within 2 seconds. 2. Form validation must execute in real time, highlighting errors before submission. 3. The account creation and activation process must complete within 3 seconds of a valid form submission. 4. All submitted registration data must be stored securely and encrypted where applicable.
Postconditions	1. A new Buyer/Tenant account is created and immediately active. 2. The user is redirected to the Buyer/Tenant dashboard.

UC_32: Browse and filter listings

UC Name	Browse and filter listings
Dependency	None
Actors	Buyer/Tenant (logged in or not)
Preconditions	1. The platform is accessible and operational. 2. At least one active listing exists on the platform.

Description of Main Sequence	1. User navigates to the TrustEstate platform - no login required. 2. System displays all publicly active listings. 3. User applies filters: location, price range, property type, and listing type (sale/rent). 4. System displays all listings matching the selected filters. 5. User browses the results and selects a listing to view its details. 6. If the user attempts an action requiring login (e.g. saving to favourites, submitting an offer): System prompts the user to log in.
Description of Alternative Sequence	1. No listings match the selected filters: System displays a message indicating no results were found and suggests adjusting the filters. 2. No active listings on the platform: System displays a message indicating no listings are currently available.
Non-functional requirements	1. The listings page must load within 3 seconds without requiring a login. 2. Filter results must update within 2 seconds of the user applying or changing a filter. 3. The system must ensure that only Active listings are displayed to unauthenticated users. 4. The platform must support concurrent browsing by multiple users without performance degradation.
Postconditions	1. User has browsed and filtered the available listings. 2. No changes to any listing are made by this action.

UC_33: View listing details

UC Name	View listings details
Dependency	UC_32 (Browse and filter listings
Actors	Buyer/Tenant (logged in or not)
Preconditions	1. The platform is accessible and operational. 2. The user has selected a listing from the browse results.
Description of Main Sequence	1. User selects a listing from the browse results. 2. System displays the full listing page including photos, title, description, location, price, listing type, current status, and the assigned Agent's profile. 3. User reviews the listing information. 4. If the listing is Active and the user is logged in: User may save the listing to favourites or submit an offer. 5. If the listing is not Active: System informs the user of its current status.
Description of Alternative Sequence	1. Listing no longer available: System displays a message indicating the listing is no longer active or has been removed. 2. User not logged in and attempts a restricted action: System prompts the user to log in.

Non-functional requirements	1. The listing detail page must load within 3 seconds. 2. All listing information displayed must reflect the most current data stored in the system. 3. The system must ensure that restricted actions such as saving to favourites or submitting an offer are only available to authenticated users.
Postconditions	1. User has viewed the full details of the selected listing. 2. No changes to the listing are made by this action.

UC_34: Save favorite listings

UC Name	Save favorite listings
Dependency	US_01 (User logs in) US_33 (View listing details)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. 2. The listing is active and publicly visible.
Description of Main Sequence	1. User navigates to a listing detail page. 2. User clicks the Save to Favourites button. 3. System adds the listing to the user's personal favourites list. 4. User navigates to the Favourites section from the dashboard to view all saved listings. 5. User may remove any listing from their favourites at any time.
Description of Alternative Sequence	1. Listing already saved: System displays a message indicating the listing is already in the user's favourites.
Non-functional requirements	1. The save action must be processed and confirmed to the user within 2 seconds. 2. The Favourites section must load within 3 seconds. 3. The system must ensure that the favourites list is private and accessible only to the authenticated user who owns it.
Postconditions	1. The listing is added to the user's personal favourites list and is accessible from the Favourites section of the dashboard.

UC_35: Submit offer

UC Name	Submit offer
Dependency	US_01 (User logs in) US_33 (View listing details)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. 2. The listing is Active. 3. The user does not already have an active offer on the same listing.

Description of Main Sequence	1. User navigates to an Active listing's detail page and clicks Submit Offer. 2. User enters a proposed price and an optional message to the Agent. 3. System checks that the user does not already have an active offer on this listing. 4. User confirms and submits the offer. 5. System creates the offer with a Pending status and notifies the assigned Agent and Property Owner. 6. User receives a confirmation that their offer has been submitted successfully.
Description of Alternative Sequence	1. User already has an active offer on the listing: System prevents submission and displays a message informing the user they already have an active offer on this listing. 2. Listing is no longer Active at time of submission: System prevents submission and informs the user of the listing's current status.
Non-functional requirements	1. The offer submission must be processed and confirmed within 3 seconds of the user's confirmation. 2. Notifications to the assigned Agent and Property Owner must be delivered within 1 minute of the offer submission. 3. The system must enforce the one-active-offer-per-listing constraint in real time, preventing duplicate active offers automatically. 4. All offer submissions must be logged with a timestamp for audit and dispute resolution purposes.
Postconditions	1. The offer is created with a Pending status. 2. The assigned Agent and Property Owner are notified of the new offer.

UC_36: Participate in negotiation

UC Name	Participate in negotiation
Dependency	US_01 (User logs in) US_33 (View listing details) US_19 (Manage offer lifecycle)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. 2. The Agent has responded to the user's submitted offer with an acceptance, decline, or counter.
Description of Main Sequence	1. User receives a notification that their offer has received a response from the Agent. 2. If the offer is accepted: User is notified; the listing moves to Under Offer status and the user awaits inspection scheduling. 3. If the offer is declined: User is notified and may submit a new offer if desired. 4. If the offer is countered: User reviews the Agent's revised price and response deadline. 5. User selects an action: Accept the counter-offer, Decline it, or Submit a revised offer. 6. System records the response and notifies the Agent. 7. Negotiation continues for a maximum of 3 rounds; if unresolved, the offer expires and the listing returns to Active status.

Description of Alternative Sequence	1. User does not respond within the response deadline set by the Agent: The counter-offer expires and the listing returns to Active status.
Non-functional requirements	1. Offer response notifications to the Buyer/Tenant must be delivered within 1 minute of the Agent's action. 2. The system must enforce the maximum of 3 negotiation rounds automatically, expiring the offer without requiring manual intervention. 3. The response deadline set by the Agent must be enforced by the system automatically upon expiry. 4. All negotiation actions must be logged with timestamps for audit and dispute resolution purposes.
Postconditions	1. If accepted: Offer status is Accepted and listing moves to Under Offer. 2. If declined: Offer is closed. 3. If revised offer submitted: Negotiation round count increments and the Agent is notified. 4. If 3 rounds reached without resolution: Offer expires and listing returns to Active status.

UC_37: Withdraw offer

UC Name	Withdraw offer
Dependency	US_01 (User logs in) US_35 (Submit offer)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. 2. The user has an active offer on a listing that has not yet been accepted.
Description of Main Sequence	1. User navigates to their active offers from the dashboard. 2. User selects the offer they wish to withdraw. 3. System verifies that the offer has not yet been accepted. 4. User confirms the withdrawal. 5. System updates the offer status to Withdrawn and notifies the assigned Agent.
Description of Alternative Sequence	1. Offer has already been accepted: System prevents withdrawal and displays a message informing the user that an accepted offer cannot be withdrawn through this action.
Non-functional requirements	1. The withdrawal must be processed and reflected in the system within 3 seconds of the user's confirmation. 2. The notification to the assigned Agent must be delivered within 1 minute of the withdrawal. 3. The system must enforce that accepted offers cannot be withdrawn through this flow — the restriction must be applied automatically.
Postconditions	1. The offer status is updated to Withdrawn. 2. The assigned Agent is notified of the withdrawal.

UC_38: View inspection report (Buyer/Tenant)

UC Name	View inspection report
Dependency	US_01 (User logs in) US_13 (Assign final verdict)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. An inspection report has been submitted for a listing on which the user has an accepted offer.
Description of Main Sequence	1. receives a notification that the inspection report is available for their accepted offer listing. 2. User logs in following UC_01 and navigates to the relevant listing. 3. System displays the full inspection report including category findings (Structural Integrity, Plumbing, Electrical, Safety), pass/fail ratings, severity classifications, supporting photos, and the final verdict. 4. User reviews the report findings and the final verdict. 5. Based on the verdict, user may choose to proceed with the transaction or withdraw/revise their offer.
Description of Alternative Sequence	1. User attempts to view a report for a listing on which they do not have an accepted offer: System prevents access and displays a message indicating they are not authorized to view this report.
Non-functional requirements	1. The inspection report must load and display in full within 3 seconds of navigation. 2. The notification to the Buyer/Tenant upon report availability must be delivered within 1 minute of the Inspector's verdict submission. 3. The system must enforce that only Buyer/Tenants with an accepted offer on the relevant listing can access the inspection report. 4. The report must be displayed in a clear and structured format organized by the four inspection categories.
Postconditions	1. User has reviewed the full inspection report. Based on the verdict, the user may proceed to UC_39 if they wish to withdraw or revise their offer.

UC_39: Withdraw or revise offer after inspection

UC Name	Withdraw or revise offer after inspection
Dependency	US_01 (User logs in) US_38 (View inspection report)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. 2. The inspection verdict is Failed or Passed with Conditions. 3. The user is within the defined response window following the verdict.

Description of Main Sequence	1. User has reviewed the inspection report and the verdict is Failed or Passed with Conditions. 2. System notifies the user of the available options within the defined response window. 3. If the verdict is Failed: User may withdraw the offer; system updates the offer status and notifies the Agent and Property Owner. 4. If the verdict is Failed or Passed with Conditions: User may submit a revised offer to renegotiate the price. 5. User selects their preferred action and confirms. 6. System processes the action, updates the offer and listing status accordingly, and notifies all relevant parties.
Description of Alternative Sequence	1. Response window has expired: System prevents the action and displays a message indicating the response window has closed. 2. Verdict is Passed: System does not present withdrawal or revision options as the inspection outcome requires no renegotiation.
Non-functional requirements	1. The available post-inspection actions must be presented to the user within 1 minute of the verdict notification. 2. The system must enforce the defined response window automatically, preventing actions once the window has expired. 3. Any withdrawal or revised offer submission must be processed and reflected in the system within 3 seconds of the user's confirmation. 4. Notifications to all relevant parties following the user's action must be delivered within 1 minute.
Postconditions	1. If withdrawn: Offer status is updated and the listing may return to Active status. 2. Agent and Property Owner are notified. 3. If revised offer submitted: The new offer enters the negotiation workflow and the Agent is notified.

UC_40: Message Agent

UC Name	Message Agent
Dependency	US_01 (User logs in) US_35 (Submit offer)
Actors	Buyer/Tenant
Preconditions	1. Buyer/Tenant is authenticated. 2. The user has submitted an offer on at least one listing, establishing a messaging relationship with the assigned Agent.
Description of Main Sequence	1. User navigates to the messaging section of the platform. 2. User selects the Agent assigned to a listing on which they have submitted an offer. 3. User composes and sends a message through the platform's internal messaging system. 4. System delivers the message and notifies the Agent. 5. Agent may reply through the same messaging thread.

Description of Alternative Sequence	1. No eligible Agent to message: System displays a message indicating the user has not submitted any offers and therefore has no active messaging relationships with an Agent.
Non-functional requirements	1. Messages must be delivered to the Agent within 1 minute of being sent. 2. The messaging interface must load within 2 seconds. 3. The system must ensure that Buyer/Tenants can only initiate or participate in messaging threads with Agents assigned to listings on which they have submitted an offer. 4. All messages must be stored securely and retained for audit and dispute resolution purposes.
Postconditions	1. The message is delivered to the assigned Agent through the platform's internal messaging system. 2. The Agent is notified.